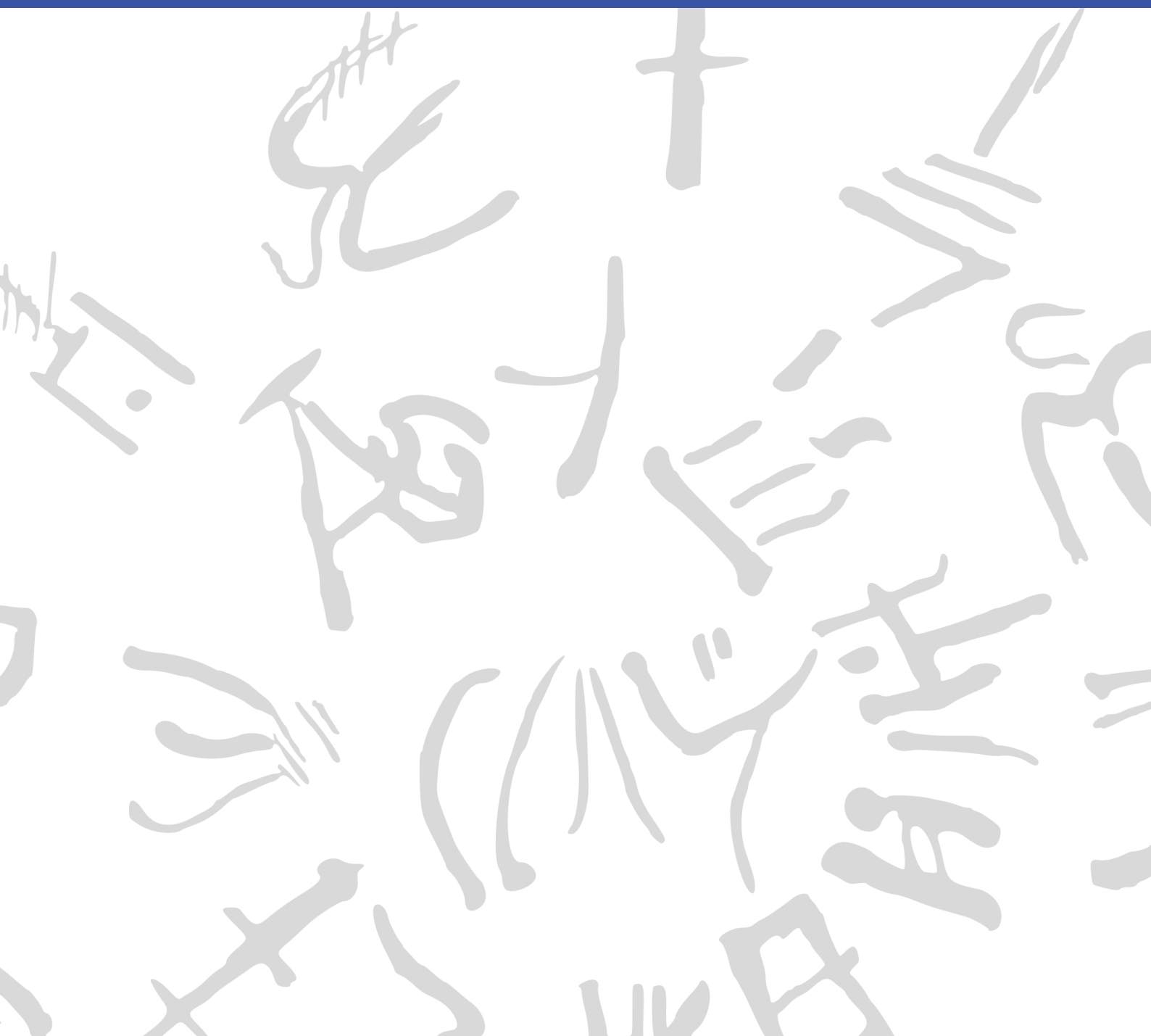


Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

Bachelor Thesis



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June, 2026

By
Frederik W. L. Christoffersen

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Approval

This thesis has been prepared over four months at the Section for Cognitive Systems, Department of Applied Mathematics and Computer Science, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Bachelor of Science in Engineering, BSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Frederik W. L. Christoffersen - s234644

.....
Signature

.....
Date

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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Author of this thesis.

Morten Mørup, Professor, DTU Compute
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Co-supervisor, external collaborator, and creator of SigLA.

Lukas Esterle, Associate Professor, Aarhus University
Co-supervisor.

Generative AI tools were occasionally used as support for brainstorming, linguistic editing, code exploration, and research during the development of this project.

Contents

Preface	ii
Abstract	iii
Acknowledgements	iv
List of tables and figures	vi
1 Introduction	1
1.1 A problem of manual digitization	1
1.2 A problem of copyright	2
1.3 Research questions	2
1.4 Outline of the thesis	2
2 Methodology	3
2.1 Segment Anything Model	3
2.2 Evaluation Metrics	3
3 Implementation	5
4 Data	7
5 Results	9
6 Discussion	11
7 Conclusion	13
8 Future Work	15
Bibliography	16
A Appendix	17

List of tables and figures

1 Introduction

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, currently consisting of only around 1,500 documents - significantly smaller than that of its younger counterpart, which now spans more than 6,000 documents.¹ But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.² Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael), these are only available in print form.

This motivated the development of the palaeographical database *The Signs of Linear A* (SigLA), which seeks to "[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions" (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known documents.^{3,4}

1.1 A problem of manual digitization

Nowadays, when archaeologists digitize ancient inscriptions, they still tend to do so manually using no more than painting tools and their own good judgement. The process is slow, and requires manual tracing of sign engravings from stone, clay, and other materials with damaged and irregular surfaces. This is perhaps the biggest reason to why SigLA does not yet contain the entire available corpus. The drawings currently stored on SigLA were made by Ester Salgarella. For each document, she created her own annotated drawings based on the originals from Cefael. According to her, this has been a highly time-consuming undertaking.

The aim of this project is therefore to investigate whether semi-automatic segmentation using modern computer vision - specifically the Segment Anything Model (SAM) - can accelerate this workflow while maintaining high-quality outputs that are still compatible with SigLA's database.

The thesis will develop an annotation tool that can generate initial segmentations from images of original documents and export cleaned masks in researcher-friendly formats (e.g., vector or layer based files), leaving only simple user corrections to the researchers.

¹Steele 2023.

²Consiglio Nazionale delle Ricerche 2025.

³SigLA 2021a.

⁴SigLA 2021b.

1.2 A problem of copyright

Due to copyright issues, SigLA cannot use the accompanying drawings for each document present provided by Cefael (collections de l'école française d'Athènes en ligne). Therefore, they have so far made their own drawings based on the images of the original documents. Consequently, SigLA's digitization workflow could benefit significantly from a tool that can enhance this process by automatically producing initial segmentation proposals from photographs of the original inscriptions. Researchers could then proceed to correct these proposals instead of tracing every sign from scratch for every new document to be digitized.

1.3 Research questions

1.4 Outline of the thesis

2 Methodology

2.1 Segment Anything Model

2.2 Evaluation Metrics

3 Implementation

4 Data

5 Results

6 Discussion

7 Conclusion

8 Future Work

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A Appendix

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